JEFF HANNA

Seattle, WA | LinkedIn

Security Engineer with 10+ years of experience who approaches every environment as a problem to be fully understood — driven by relentless curiosity to trace anomalies to root cause, and the engineering discipline to build automated, AI-augmented systems that close detection and response gaps permanently.

CORE COMPETENCIES

Detection Engineering (YARA, Sigma, Splunk SPL, SentinelOne) | Incident Response & Triage | Threat Intelligence Integration | SIEM/EDR/SOAR Engineering | Pipeline Architecture & Automation | LLM-Augmented Security Tooling | Python, GoLang, Bash, PowerShell | AWS (CDK, S3, Athena, EC2, IAM, ECS) | Kubernetes Security | Threat Hunting | DNS & Network Analysis | Cloud-Native Security | Containerization (Docker/Podman/Rancher) | Program Development & Stakeholder Management

KEY ACCOMPLISHMENTS

- Engineered an LLM-orchestrated threat intelligence pipeline from the ground up that ingests raw security bulletins and produces tiered, audience-specific intelligence: executive summaries, defensive recommendations, and SIEM/EDR/endpoint detection rules with hunt queries. A custom normalization and tag-filtering layer reduces input volume by over 99% (millions of tokens to ~10k). An agentic enrichment loop queries threat intelligence APIs across threat actors, malware families, vulnerabilities, and campaigns to ground synthesis in live data. Per-phase token instrumentation enables diagnosis of reasoning quality and context degradation across pipeline stages. Built using a multi-model architecture targeting AWS infrastructure with full SDLC and git-based development workflow.
- Conducted a DNS intelligence experiment analyzing ~1 billion packets (~170GB across ~2,000 compressed files) to evaluate DNS as a detection source — and confirmed previously undetected threat activity in the hosting environment. Analysis surfaced SQL injection and active pen testing attempts embedded in DNS requests, clear typosquatting patterns in non-Latin character sets, and other anomalous behaviors not identified by any other detection system. Built Python algorithms assessing DGA patterns, entropy and character anomalies, homograph/punycode risks, TTL outliers, and transaction ID reuse across 31 million analyzed domains. Designed and iterated AWS infrastructure (CDK, S3, Athena, EC2, ECS) and containerized processing pipelines to enable multi-threaded analysis at scale. Produced the strategic roadmap and technical requirements for a 2026 production DNS detection program.
- Engineered automated employee credential remediation by building Splunk correlation searches and SOAR integrations that ingest compromised credential data, validate username/password pairs against live systems, and auto-rotate confirmed exposures — processing 793 inputs, rotating 16 true exposures, eliminating 302 false positive rotations, and saving 132+ hours of annual manual effort. Separately served as program owner for the customer credential remediation program, designing solution architecture and coordinating AWS infrastructure delivery with a partner engineering team.
- Proactively identified a silent authentication failure in a production system outside my ownership — independently pulled the repository, navigated CloudWatch using logger names to correlate log entries, and traced incorrect output for one of two pipeline groups to an expired JWT token. Surfaced the root cause to the owning team, who resolved the credential and implemented CloudWatch alerting to prevent recurrence.
- Built a novel SOC-style alert triage workflow inside ThreatStream using threat bulletins and a custom tagging system — an approach so unconventional that Anomali customer success managers had never seen it implemented. Automated a secondary filtering layer that reduced daily alert volume by 73% (52 → 14/day). The system processed 3,315 alerts over eight months and escalated 109 actionable items to security teams that would otherwise have been missed entirely.
- Integrated Anomali ThreatStream IoC feeds into Splunk SIEM and SentinelOne EDR; established confidence scoring, correlation logic, dashboards, and ongoing source quality tuning. The integration delivered 1,500+ notable events to the SOC, contributed to blocking millions of malicious connection attempts, and produced 24 true positive incidents requiring remediation. Identified and restored a months-long EDR integration gap. Feed quality improvements positioned the detections team to lower confidence thresholds in 2026 for broader threat coverage.
- Developed a CTI workflow adopted by a partner organization that reduced their average resolution time by 57% for infringement notice delivery and will generate $350k in annual savings starting in 2026 through elimination of a redundant SaaS subscription.

PROFESSIONAL WORK EXPERIENCE

Senior Security Engineer | GoDaddy 03/2022 – Present
Security Program Engineering 70% | Detection & Response 15% | Threat Intelligence 10% | Threat Hunting 5%

- Architect and build automated security programs end-to-end: from requirements definition through infrastructure design, detection logic, SOAR automation, and operational dashboarding — often as sole engineer on greenfield initiatives.
- Design and deploy LLM-augmented security tooling, including an agentic threat intelligence pipeline with multi-model orchestration, enrichment loops against live threat intel APIs, and tiered report generation for both executive and engineering audiences.
- Author and maintain Splunk detections and SentinelOne rules from open to close; coordinate with detections engineering for peer review on complex scheduled searches and memory-intensive logic.
- Built and operated a DNS intelligence capability processing ~1 billion packets using Python, Docker, and AWS (CDK, S3, Athena, EC2); developed anomaly detection algorithms for DGA, entropy, homograph/punycode, TTL, and transaction ID patterns across 31 million analyzed domains; confirmed detection gaps in the hosting environment through findings not surfaced by any other security system.
- Engineered automated credential remediation workflows (employee and customer) using Splunk, SOAR, and AWS; discovered and diagnosed a week-long production outage in the customer program through active monitoring of self-built metrics dashboards.
- Identified a production authentication failure in an external team's system by independently reading their codebase, tracing CloudWatch logs via logger names, and pinpointing an expired JWT causing silent failures — surfaced root cause and drove resolution.
- Designed and implemented ThreatStream-to-Splunk and ThreatStream-to-SentinelOne integrations; replaced personal auth tokens with service accounts, tuned IoC source quality, and maintained continuous operation throughout.
- Built a novel alert triage system using ThreatStream capabilities in unintended ways; engineered automated pre-filtering that reduced queue volume 73%; processed 3,315 alerts with 109 escalations in eight months.
- Supported 30+ security incidents as CTI/IR collaborator; proactively initiated 7 incidents through intelligence discovery including exposed GitHub repositories and credential leaks.
- Served as acting senior manager for one quarter: led midyear reviews, ran leadership syncs, coordinated IR/TH/VM collaboration, and maintained personal engineering output — selected over more senior peers.
- Briefed quarterly threat intelligence reports to 100+ security personnel at InfoSec all-hands; product used by CISO as primary security posture touchpoint.
- Conducted Anomali Copilot PoC; rigorous testing and detailed engineering feedback led Anomali to provide the tool to GoDaddy at no cost ($20k/year value) and elevated the organization to Anomali VIP customer status.
- Mentored principal engineers on AWS CDK, IAM, and cloud-native development; unblocked two separate projects and saved estimated days-to-weeks of trial and error.

Intelligence Directorate Branch Supervisor | U.S. Cyber Command 12/2017 – 10/2021
Program Development 40% | Personnel Management 25% | Technical Analysis 25% | Training 10%

- Developed USCYBERCOM's first OSINT program: established processes, tooling, governance, and data quality standards from scratch; managed publication of 500+ technical products consumed by USCYBERCOM Commander, NSA Director, and the President of the United States.
- Built threat hunt packages supporting Hunt Forward Operations (HFOs) using MITRE ATT&CK, Diamond Model, and Cyber Kill Chain frameworks; managed and trained a team of 6 cyber intelligence analysts.
- Created and maintained large-scale data processing workflows organizing 500,000+ data points for executive decision-making; partnered with government and commercial entities to automate OSINT acquisition and enrichment.

Intelligence and Cyber Programs Manager | Washington Army National Guard 03/2015 – 10/2022
Program Development 70% | Training & Education 10% | Executive Briefing 10% | Technical Production 10%

- Built core CTI and information security programs from scratch including PIR frameworks, collection and dissemination workflows, and partner-facing data management structures used across multiple organizations.
- Led and mentored up to 50 full- and part-time personnel; partnered with national and international stakeholders to align capabilities and build enduring collaborative programs.
- Partnered with Defensive Cyber Operations (DCO) to build redundant defensive capabilities and establish policy and training pipelines.

EDUCATION & PROFESSIONAL DEVELOPMENT

Master of Cybersecurity and Leadership – University of Washington (GPA: 3.85)
Cybersecurity Engineer Graduate Certificate – SANS Technology Institute (2024)
Cloud Security Graduate Certificate – SANS Technology Institute (Est. 2026)

SANS Training:
SEC595 Applied Data Science and AI/ML for Cybersecurity (Est. 2026) | SEC540 Cloud Native Security & DevSecOps Automation | SEC510 Cloud Security Controls & Mitigations (2025) | SEC488 Cloud Security Essentials (2024) | SEC503 Network Monitoring & Threat Detection (2024) | SEC504 Hacker Tools, Techniques & Incident Handling (2024) | SEC401 Security Essentials (2023) | FOR578 Cyber Threat Intelligence (2022)

Certifications:
CISSP | GSEC | GCIH | GCIA | GCTI | GCSA | GPCS | GCLD | CMMC CCP

PROFESSIONAL ORGANIZATIONS

Project Management Institute (PMI) | Women in Cybersecurity (WiCyS) | CompTIA | National Diversity Council | National Guard Association of Washington (NGAW)